JANUARY - JUNE 2023

GOVERNMENT ENGINEERING COLLEGE PATAN

NEWSLETTER

INSTITUTE GLIMPSE

Government Engineering College, Patan (G.E.C. Patan), situated in the tranquil and scenic village of Katpur in Patan District, Gujarat, was established in 2004 with a commitment to delivering quality education in engineering and technology. Over the years, the institute has achieved remarkable growth and excellence in academics, innovation, and research. It is affiliated with the Gujarat Technological University (GTU) and functions under the administration of the Directorate of Technical Education (DTE), Government of Gujarat.

INSTITUTE VISION

"To prepare human resources with value-based competency for technical advancements and growth of society."

INSTITUTE MISSION

- **To Deliver technical programs and services to cater the current needs of society and industry.**
- **Helping industries in solving challenges by means of providing best technical human resources.**
- **To contribute in sustainable growth of society.**



ELECTRONICS AND COMMUNICATION DEPARTMENT

NEWSLETTER

ABOUT

The Electronics and Communication Engineering (ECE) Department at Government Engineering College, Patan, established in 2004, offers a four-year undergraduate program with modern infrastructure, advanced laboratories, and experienced faculty. The department provides strong foundations in electronics, communication, networking, wireless technologies, and VLSI, preparing students for careers, research, and higher education.

DEPARTMENT VISION

"To prepare engineers with essential technical knowledge, value-based professional skills for technical upgradation and societal growth."

DEPARTMENT MISSION

- **To prepare competent engineers with the ability t› s›lve in ustrial problems who can raise the living stan ards of the society.**
- **To prepare engineers with effective communication skills, professi›nal ethi›s and lea ership qualities.**



NEWSLETTER

INDEX

Title	Page No.
Institute Vision Mission	1
Department Vision Mission	2
INDEX	3
HOD's Message About Department	4
PEOs - PSOs	5
Faculty Details	6
Department Facilities	7 to 8
Department Activities	9 to 10
Students Activities	11
Committee Activities (NSS)	12 to 18

HOD'S MESSAGE



PROF. SANJAY D. JOSHI

It is my immense pleasure to present this Newsletter of the Electronics and Communication Engineering Department, Government Engineering College, Patan — a comprehensive documentation of our academic journey, industry exposure, and innovation across the years.

This edition reflects the relentless dedication of our faculty and the enthusiasm of our students. From impactful industry visits and expert lectures to project presentations and faculty development programmes, each activity represents our collective commitment to bridging academia and industry. I express my heartfelt gratitude to all stakeholders — our students, faculty, industry partners, and the college administration — who have continually contributed to making our department a centre of excellence.

~

PROF. SANJAY D. JOSHI
HOD EC Department
Government Engineering College, Patan

ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- Accumulate sound technical knowledge that enables to pursue higher education and research.
- Work in multidisciplinary areas with a strong focus on innovation and entrepreneurship.
- To inculcate effective communication skills, managerial skills, as well as professional ethics for successful professional career.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- An ability to apply the conceptual knowledge of Microelectronics, Signal Processing, Microprocessors/Microcontrollers and Communication Systems to solve industrial problems.
- An ability to solve Electronics and communication Engineering problems using the latest hardware and software.

ELECTRONICS AND COMMUNICATION ENGINEERING

Faculty Members



PROF. SANJAY D. JOSHI
DESIGNATION: ASSOCIATE PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM, IMAGE PROCESSING, MICROPROCESSOR - MICROCONTROLLER



DR. HARESHKUMAR JUDAL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DIGITAL COMMUNICATION,WIRELESS COMMUNICATION AND DSP



PROF. MIHIRKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DSP, COMMUNICATION SYSTEMS



DR. MAULIKKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: MICROWAVE, ANTENNA, ELECTRONICS



PROF. JAYESHKUMAR C. PRAJAPATI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: COMMUNICATION ENGG, MICROWAVE ENGG, OPTICAL FIBER COMMUNICATION



PROF.SHYAMKUMAR PANKHANIYA
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM DESIGN, VLSI TECHNOLOGY



PROF. MEHULKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: VLSI DESIGN



PROF. ROSHNIBEN CHAUDHARI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: ANTENNA, RF & MICROWAVE ENGINEERING, WIRELESS COMMUNICATION, BASIC ELECTRONICS



PROF. ANJU VASDEWANI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEMS, DIGITAL CIRCUITS AND DESIGNING, MICROPROCESSOR, MICRO CONTROLLER & INTERFACING, PROGRAMMING USING C, BASIC ELECTRONICS

ELECTRONICS AND COMMUNICATION ENGINEERING

FACILITIES OF EC DEPARTMENT

L201 ANALOG & DIGITAL ELECTRONICS LABORATORY

This laboratory is equipped with Analog and Digital Electronics trainers, viz.,amplifier, common emitter configuration, common collector configuration,oscillators, multivibrators, power supply, trainers using 555 IC, digital gates, flip flops, counters, and registers kits, etc. Following are few kits:

- Active band pass filter
- Automatic gain control kit
- Butterworth filter trainer
- CE configuration of transistor
- DC motor speed control using SCR



L202 SIMULATION LABORATORY

This laboratory is equipped with the latest computers with advanced simulation software, viz., MATLAB, Multisim, Visual Studio Code, Keil uVision, Atmel Studio and Xilinx etc. Laboratory is equipped with:

- DIGITAL OSCILLOSCOPE DSOX 2024A
- Workbench Electronics



L-203 ELECTRONIC SYSTEM DESIGN LABORATORY

This laboratory consists of multiple electronics workbenches, having a digital storage oscilloscope, multi meter, function generator, and power supply, which can be helpful in testing and measurements of various electronics circuits or systems.



ELECTRONICS AND COMMUNICATION ENGINEERING

FACILITIES OF EC DEPARTMENT

L211 COMMUNICATION LABORATORY

This laboratory is equipped with Analog and Digital communication trainers, viz., AM, FM, and PM modulation demodulation, superheterodyne receiver, PWM, PPM, PAM, ASK, FSK, PSK kits, Spectrum Analyzer as well as RF and Microwave trainers for microwave communication along with active and passive components. Few trainers are as:

- Amplitude Modulation/Demodulation Trainer
- DSB/SSB AM Receiver
- Fiber Optics Demonstrator
- FM Transmitter Receiver Trainer
- Microwave Bench



L212 PROJECT LABORATORY

This laboratory is designed to provide students with hands-on experience in research and practical applications across various fields of Electronics and Communication Engineering.



L213 SIGNAL PROCESSING & VLSI LABORATORY

This laboratory consists of various trainers, viz., DSP, FPGA kits, and software required to perform operations on the trainers for Digital Signal Processing, Image Processing, Speech Signal Processing,
and VLSI experiments.

- DSP Trainer Kit
- Desktop Computer
  (ACER- AMD processor 12th generation)
- Programmable Function Generator, 1 Channel, 50Mhz (AFG3051C)



ELECTRONICS AND COMMUNICATION ENGINEERING

Impact Lecture on "Achieving Problem Solution Fit & Product Market Fit"

Date: 24th February 2023
Time: 11:00 A.M. onwards
Venue: Google Meet (Virtual)



Institution Innovation Council (IIC) an Initiative of MHRD for selected Higher Education Institutes. Ministry of Human Resource Development (MHRD), Govt. of India has established 'MHRD's Innovation Cell (MIC)' to systematically foster the culture of Innovation amongst all Higher Education Institutions (HEIs).

Electronics and Communication Department has organised Impact Lecture on "Achieving Problem Solution Fit & Product Market Fit" by an Industrial Expert Mr. Niravgiri Goswami Founder and Director " of "i- SPARSH Technology" which is a start up funded by Government of Gujarat and incubated at PDPU University Gandhinagar.

We have organised program at 24th February 2023 Friday in the time duration of 11:00AM to 12:30 P.M. by virtual mode of Google Meet. We have invited the student from all department and around 50 registered and participated in the program. Speaker has given a so much valuable tips to student regarding student start-up program as well as Intellectual Property Right . Students have interacted with the speaker eagerly and encouraged to do the needful in the area of Start-up as well as Intellectual Property Right.

ELECTRONICS AND COMMUNICATION ENGINEERING

Industrial Visit

Date: 26th April 2023

Industry Visit on 26/04/2023 for 4th and 6th sem EC/CSE students.

Morning: Teksun Microsysy Ltd., Ahmedabad (An Embedded System and IoT service)

Afternoon: AlphaNumero, Ahmedabad (VLSI/Chip Verification service)













ELECTRONICS AND COMMUNICATION ENGINEERING

Date: 02nd May 2023

Event Place: Google meet (Virtual)

Electronics and Communication Department students have participated in Impact Lecture on "Business Model Canvas " by an Expert Dr. R.Mohana Priya Associate Professor of ECE Department , IIC Convener and Innovation Ambassador, Pavaai Engineering College, Pachal , Nammakal District.

This program was organized on 2ND May 2023 Tuesday in the time duration of 6:30 P.M. to 7:30 P.M. by virtual mode of Google Meet. Speaker has given a so much valuable tips to student regarding Business Model Canvas , student start-up program as well as Legal and Ethical Steps . Students have interacted with the speaker eagerly and encouraged to do the needful in the area of Business Model Canvas and Start-up as well as Legal and Ethical issues.







NATIONAL YOUTH DAY

Event date : 12th January 2023

National Youth Day, also known as Swami Vivekanand Jayanti, is celebrated in India on January 12 every year to commemorate the birth anniversary of Swami Vivekanand one of India's most influential and spiritual leader and thinkers.

As on this NSS GEC Patan Organized an event in which Principal GEC Patan along with Faculties from various departments were present. The event started with prayer song followed by presentation in which Students were motivated with Spiritual and Energetic thoughts of Swami Vivekananda by Vikhnesh Pillai an NSS Volunteer. After that An Essay writing competition along with poster making competition was organized by NSS Family in order to upbring the hidden talents among the youth of India. Also the faculties Of college shared their views on todays youth and tried to motivate all of us.

One of the faculty also shared Chicago incidence of swami Vivekananda that how he started his journey as a young influencer and influenced the world leaders about importance of religion and spirituality in life which also influenced the todays youth and we all were motivated.

The winners of both the competitions were encouraged by principal by giving the Prize to winners. The winner of Essay Competition was URJA KORIYA student from CSE Department and the winner of Poster making Competition was TRUSHI another Student from CSE Department. At last on concluding the event all the audience sang National Anthem and event was concluded with positivity.

Glimpse of the Event



પાટણની સરકારી ઈજનેરી કોલેજ ખાતે આજે રાષ્ટ્રીય યુવા દિવસની ઉજવણી કરાઈ

(દિનેશ) પાટણ, પાટણની સરકારી ઈજનેરી કોલેજ ખાતે આજે રાષ્ટ્રીય યુવા દિવસની ઉજવણી કરાઈ હતી.

૧૨ જાન્યુઆરી - ૧૮૬૩ ના રોજ યુવાનો ના પ્રેરણાસ્ત્રોત સ્વામી વિવેકાનંદ નો જન્મ થયો હતો. જેમના જીવનના વિવિધ પાસા યુવાનો માટે પ્રેરણા રૂપ બની રહે તે હેતુથી ૧૨ જાન્યુઆરી રાષ્ટ્રીય યુવા દિવસની ઉજવણી કરવામાં આવી હતી. જેના ભાગરૂપે આજ રોજ કોલેજ ખાતે વિવિધ કાર્યક્રમો યોજાયા હતા. જેમાં સૌ પ્રથમ કોલેજના વિદ્યાર્થીઓ માટે નિબંધ સ્પર્ધા યોજાઈ હતી. જેમાં ૧૮૦ વિદ્યાર્થીઓએ ભાગ લીધો હતો. આ સ્પર્ધા ના વિજેતા થયેલા વિદ્યાર્થીઓને ઈનામ આપીને









Seminar On G20 Presidency

Event Date : 21st January 2023

The Group of Twenty (G20) is the premier forum for international economic cooperation. It plays an important role in shaping and strengthening global architecture and governance on all major international economic issues. The Group of Twenty (G20) comprises 19 countries (Argentina, Australia, Brazil, Canada, China, France, Germany, India, Indonesia, Italy, Japan, Republic of Korea, Mexico, Russia, Saudi Arabia, South Africa, Türkiye, United Kingdom and United States) and the European Union. .



On the occasion of the G20 summit 2023, a seminar was arranged in our college in which college faculties & NSS Volunteers are present. At the beginning of the seminar respected principal Dr. H.S. Patel give a brief note on the G20 Presidency of India. Afterward, professor Viral Pansiniya & D.K. Patel sir also shares the objectives of G20 through a Presentation. The work and management of NSS Volunteers throughout the seminar are appreciated.







Republic Day

Event Date : 26th January 2023

Venue : GEC Patan Campus

On 26th January 2023, NSS Volunteers took the stage with full josh and dedication towards the nation to celebrate Republic Day Celebration. The celebration started with a flag-hoisting ceremony headed by our acting principal. After the flag hoisting, our GISF guard took the ground to march with full pride in their chest and head held high. After the roaring march passed by the GISF guard, NSS volunteers gave a speech on the importance of celebrating Republic Day.

At last, our acting principal took the stage to share his inspiring thoughts on Republic Day and our mission of achieving the goals of the G20 SUMMIT to lead India at the world level and making a progress in terms of the overall development of the country. He also mentioned in his speech that this is a new India facing modern problems which need modern solutions to all those problems.





Sainik Welfare Fund

Event date : 16th March 2023

As we know Soldiers are the defense backbone of our country. Their presence on the border provides a safe life to every citizen of the country. In the same way, we must give respect to them and provide help in any form. So, every year our University spread the guidelines to collect the fund to fulfill the requirements of the families of our great Soldiers.

According to the instructions & guidelines from our University, we started to collect funds named as Sainik Welfare Fund for the year 2023. We started doing collection right from the faculties of our college. Then after we do collection from local markets, residential societies, students of our college, etc. In the time of the modern era & to support digitalization the motto of the Government of India we also use the UPI payment system using a QR code for the collection.

At last, after adding all the collection of volunteers, we create a total of amount Rupees 51,151 (Fifty-one thousand one hundred fifty-one Rupees) for the Sainik Welfare Fund. On the 16th of March, 2023 we went to Gujarat Technological University, Chandkheda to submit the Demand Draft of our collection to Dr. Pankajray Patel In Charge Vice-Chancellor of GTU. As we look at last year's collection GEC Patan collects the highest sainik welfare fund in the University.



World Environment Day

Event Date : 12th June 2023



On 12th June 2023 NSS GEC Patan has organized a event for spreading awareness on issues related to environment and people roles in overcoming it following the global theme PLASTIC FREE SOCIETY Firstly the event started with Deep Pragatiya an Indian culture and pride of starting an any good event or initiative by Principal GEC Patan along with faculties. After that we moved on with a Lakshya Geet and started our first competition of Elocution with topic Youth's Role in making Plastic Free society. In which NSS Volunterrs and College Students Participated actively and shared their views and ideas infront of Audience. The results of Elocution Competition were as follows;



પાટણની સરકારી ઈજનેરી કોલેજ ખાતે વિશ્વ પર્યાવરણ દિવસની ઉજવણી કરવામાં આવી

Results Of Elocution Competition

1st RANK - Dwiwedi Srushti

2nd RANK - Sharma Dev

3rd RANK - Krisha Panchal





International Yoga Day

Event Date : 21st June 2023

Venue : GEC Patan Campus

On 21st June International Yoga Day is celebrated all over globe with the good intentions of making people of globe stress free, and fit to protect themselves from all the health related problems and protect them from negativity spread around the world. As a part of these NSS GEC Patan organised a yoga session in college campus for motivating students for performing yoga every day which would help in reducing stress level which is a major issue among students nowadays and making youth of India fit and stress free.

One of the student named Dewanshi Gauswami guided everyone with different yoga asanas and explained the importance of each asanas. In this all faculties of college including principal also joined to motivate the youth and show their interest towards fitness . We started at 6:30 and performed different asanas for around 1:30 hour also understanding the importance of each asanas.









EDITORIAL TEAM

Patron

Dr. D. H. Patel

HOD, Electronics and Communication Department

Editors-in-Chief

Dr. M. B. Patel - Electronics & Communication Engineering

Content & Design Student Team

Joshi Parv Kashyap (240220111004 - EC)

Verma Samir Santoshbhai (240220111033 - EC)

Dahivat Ahir (250022011101 - EC)

Shiv Patel (250220111023 - EC)

Harsh Prajapati (250220111025 - EC)

Rushang Thakar (250220111034 - EC)